IGCSE · Cambridge (CIE) · Maths

29 mins 14 questions

Exam Questions

Transformations of Graphs

Translations of Graphs

Easy (6 questions)	/9
Medium (4 questions)	/9
Hard (4 questions)	/11
Total Marks	**/29**

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Easy Questions

1 Find the vector that translates the point $(1, 5)$ to the point $(-1, 7)$.

(2 marks)

2 F is the point $(5, 7)$.

The vector that maps F onto the point G is $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

Find the coordinates of G.

(...................... , ...)

(1 mark)

3 P is the point $(-3, 6)$.
Q is the point $(0, 2)$.

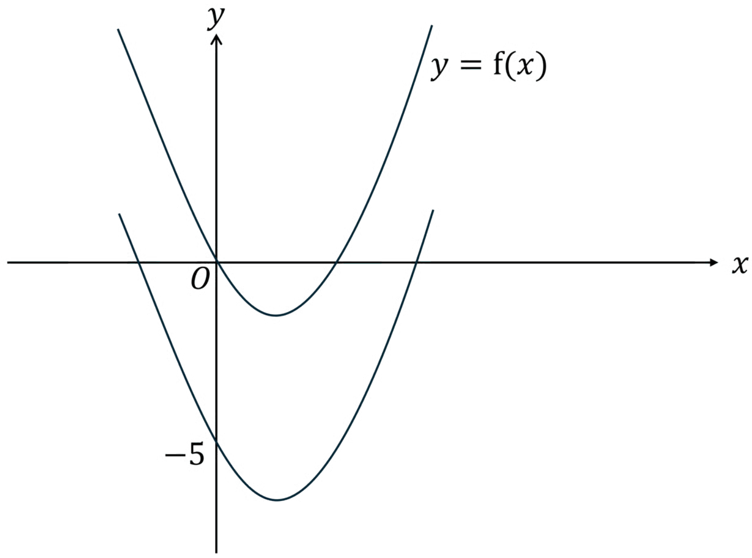
Find the translation vector that maps the point P onto the point Q.

(2 marks)

4 $(7, 13)$ is a point on the graph $y = \mathrm{f}(x)$.

Write down the coordinates of a point which must be on the graph of $y = \mathrm{f}(x) - 2$.

(1 mark)

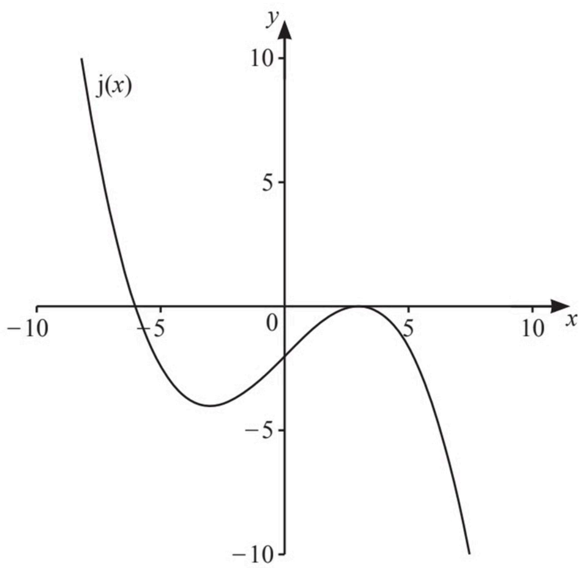


5 A curve with equation $y = f(x)$ is translated so that the point at $(0, 0)$ is mapped to the point $(0, -5)$.

Find the equation of the translated curve.

(1 mark)

6 The diagram shows a sketch of the graph of $y = j(x)$.



On the same diagram, sketch the graph of $y = j(x+2)$.

(2 marks)

Medium Questions

1 The graph with equation $y = x^2 + 2$ undergoes a translation of 3 units to the right.

Show that the equation of the translated graph simplifies to $y = x^2 - 6x + 11$.

(2 marks)

2 A curve with equation $y = \text{f}(x)$ is translated so that the point at $(0, 0)$ is mapped onto the point $(4, 0)$.

(i) Find the translation vector.

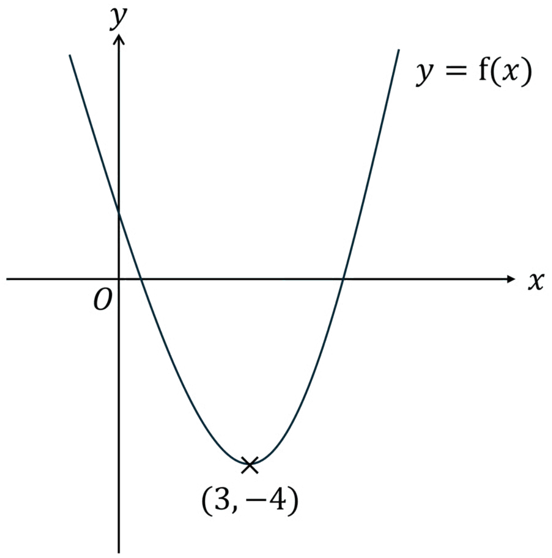
[1]

(ii) Find the equation of the translated curve.

[1]

(2 marks)

3



The diagram shows part of the curve with equation $y = \mathrm{f}(x)$.
The coordinates of the minimum point of this curve are $(3, -4)$

Write down the coordinates of the minimum point of each of the curves below:

(i) $y = \mathrm{f}(x) + 3$

[1]

(ii) $y = \mathrm{f}(x + 2)$

[1]

(2 marks)

4 The graph with equation $y = x^2 - 4x$ undergoes a translation of one unit in the negative x direction and five units in the positive y direction.

Find the equation of the translated graph.
Give your answer in the form $y = ax^2 + bx + c$.

(3 marks)

Hard Questions

1 The graph of $y = \cos(x - 35)$ for $0° \leq x \leq 360°$ crosses the x-axis at two points.

Write down the coordinates of the points where this occurs.

(2 marks)

2 (a) A curve with equation $y = \mathrm{f}(x)$ has one minimum point at $P(3, -4)$.

Write down the coordinates of the minimum point of the curve with the equation $y = \mathrm{f}(x - 5) + 6$.

(2 marks)

(b) Write down an equation of the graph if the minimum point is instead translated to the point $(0, 0)$

(2 marks)

3 The graph with equation $y = x^2 + ax + b$ undergoes a translation of two units in the positive x direction and four units in the negative y direction.

The equation of the translated graph is $y = x^2 - x - 10$.

Find the values of a and b.

(1 mark)

4 (a) The point $P(3, 2)$ lies on the curve with equation $y = \mathrm{f}(x)$.

On the graph of $y = f(x) + a$, where a is a constant, the point P is mapped to the point $(3, -5)$.

Determine the value of a.

(1 mark)

(b) On the graph of $y = f(x + b)$, where b is a constant, the point P is mapped to the point $(-1, 2)$.

Determine the value of b.

(1 mark)

(c) A different function, $g(x)$, is a cubic function of the form $g(x) = ax^3 + bx^2 + cx + d$.

The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$.

Write down the equation of the transformed graph in terms of a, b, c, d, and x.

(2 marks)